jupyter Experiment_2C_Reading_Data Last Checkpoint: 1 minute ago

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Code JupyterLab Python 3 (ipykernel)

[1]:
```
import os

print("Current folder:")
print(os.getcwd())

print("\nChecking required files...")

files = [
    "Google_data (2b.c1).csv",
    "data (2c2).xlsx"
]

for file in files:
    if os.path.exists(file):
        print("FOUND:", file)
    else:
        print("NOT FOUND:", file)
```

```
Current folder:
C:\Users\shamu

Checking required files...
NOT FOUND: Google_data (2b.c1).csv
NOT FOUND: data (2c2).xlsx
```

[2]:
```
import pandas as pd
import os

# ============================================================
# EXPERIMENT 2(C)
# Reading Data from Text Files, Excel, and Web
# ============================================================

print("=" * 65)
print("EXPERIMENT 2(C) - DATA READING AND PROCESSING")
print("=" * 65)


# ============================================================
# PART 1: CREATE REQUIRED INPUT FILES
# ============================================================

# ------------------------------------------------------------
# Create a CSV dataset similar to the Google Play Store data
# used in the laboratory manual
# ------------------------------------------------------------

csv_data = {
    "App": [
        "Photo Editor & Candy Camera & Grid & ScrapBook",
```

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Code JupyterLab Python 3 (ipykernel)

```python
csv_data = {
    "App": [
        "Photo Editor & Candy Camera & Grid & ScrapBook",
        "Coloring book moana",
        "U Launcher Lite – FREE Live Cool Themes, Hide Apps",
        "Sketch - Draw & Paint",
        "Pixel Draw - Number Art Coloring Book",
        "Paper flowers instructions",
        "Smoke Effect Photo Maker - Smoke Editor",
        "Infinite Painter",
        "Garden Coloring Book",
        "Kids Paint Free - Drawing Fun"
    ],

    "Category": [
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN",
        "ART_AND_DESIGN"
    ],

    "Rating": [
        4.1,
        3.9,
        4.7,
        4.5,
        4.3,
        None,
        3.8,
        4.1,
        4.4,
        4.7
    ],

    "Reviews": [
        159,
        967,
        87510,
        215644,
        967,
        167,
        178,
        36815,
        13791,
        121
    ],
```

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Code JupyterLab Python 3 (ipykernel)

```
    "Android Ver": [
        "4.0.3 and up",
        "4.0.3 and up",
        "4.0.3 and up",
        "4.2 and up",
        "4.4 and up",
        "2.3 and up",
        "4.0 and up",
        "4.2 and up",
        "4.0 and up",
        "4.0 and up"
    ]
}

csv_df = pd.DataFrame(csv_data)

csv_filename = "Google_data (2b.c1).csv"
csv_df.to_csv(csv_filename, index=False)

print("\nCSV input file created:")
print(csv_filename)


# ============================================================
# PART 2: CREATE EXCEL INPUT FILE
# ============================================================

excel_data = {
    "Product": [
        "Laptop",
        "Smartphone",
        "Tablet",
        "Headphones",
        "Keyboard"
    ],

    "Price": [
        1000,
        800,
        500,
        100,
        50
    ],

    "Quantity": [
        5,
        8,
        10,
        15,
        20
    ]
```

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Code JupyterLab Python 3 (ipykernel)

```
text_df.to_csv(
    processed_csv,
    index=False
)

print("Processed CSV saved successfully:")
print(processed_csv)


# ============================================================
# PART 11: SAVE PROCESSED EXCEL
# ============================================================

print("\n" + "=" * 65)
print("9. SAVING PROCESSED EXCEL")
print("=" * 65)

processed_excel = "processed_excel.xlsx"

excel_df.to_excel(
    processed_excel,
    index=False
)

print("Processed Excel saved successfully:")
print(processed_excel)


# ============================================================
# PART 12: VERIFY OUTPUT FILES
# ============================================================

print("\n" + "=" * 65)
print("10. OUTPUT FILE VERIFICATION")
print("=" * 65)

if os.path.exists(processed_csv):
    print("✓ processed_text.csv created successfully.")

if os.path.exists(processed_excel):
    print("✓ processed_excel.xlsx created successfully.")


# ============================================================
# FINAL RESULT
# ============================================================

print("\n" + "=" * 65)
print("EXPERIMENT 2(C) COMPLETED SUCCESSFULLY")
print("=" * 65)
```

Experiment_2C_Reading_Data Last Checkpoint: 2 minutes ago

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Code JupyterLab Python 3 (ipykernel)

```
================================================================
EXPERIMENT 2(C) - DATA READING AND PROCESSING
================================================================

CSV input file created:
Google_data (2b.c1).csv
Excel input file created:
data (2c2).xlsx

================================================================
1. READING DATA FROM CSV FILE
================================================================

First 5 rows of CSV data:
                                                 App        Category  Rating  \
0     Photo Editor & Candy Camera & Grid & ScrapBook  ART_AND_DESIGN     4.1
1                                Coloring book moana  ART_AND_DESIGN     3.9
2  U Launcher Lite – FREE Live Cool Themes, Hide ...  ART_AND_DESIGN     4.7
3                              Sketch - Draw & Paint  ART_AND_DESIGN     4.5
4              Pixel Draw - Number Art Coloring Book  ART_AND_DESIGN     4.3

  Reviews  Size     Installs  Type Price Content Rating  \
0     159   19M      10,000+  Free     0       Everyone
1     967   14M     500,000+  Free     0       Everyone
2   87510  8.7M   5,000,000+  Free     0       Everyone
3  215644   25M  50,000,000+  Free     0           Teen
4     967  2.8M     100,000+  Free     0       Everyone

                     Genres      Last Updated         Current Ver  \
0              Art & Design   January 7, 2018               1.0.0
1  Art & Design;Pretend Play  January 15, 2018              2.0.0
2              Art & Design    August 1, 2018               1.2.4
3              Art & Design      June 8, 2018  Varies with device
4    Art & Design;Creativity    June 26, 2018               1.0.0

   Android Ver
0  4.0.3 and up
1  4.0.3 and up
2  4.0.3 and up
3    4.2 and up
4    4.4 and up

================================================================
2. READING DATA FROM EXCEL FILE
================================================================

First 5 rows of Excel data:
      Product  Price  Quantity
0      Laptop   1000       5.0
1  Smartphone    800       8.0
2      Tablet    500       NaN
3  Headphones    100      15.0
4    Keyboard     50      20.0

================================================================
```

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Code JupyterLab Python 3 (ipykernel)

```
=================================================================
3. READING DATA FROM WEB
=================================================================

First 5 rows of Web data:
    Country  Region
0   Algeria  AFRICA
1    Angola  AFRICA
2     Benin  AFRICA
3  Botswana  AFRICA
4   Burkina  AFRICA

=================================================================
4. DATASET INFORMATION
=================================================================

CSV Dataset Shape:
(10, 13)

Excel Dataset Shape:
(5, 3)

Web Dataset Shape:
(194, 2)

=================================================================
5. CHECKING MISSING VALUES
=================================================================

CSV Missing Values:
App               0
Category          0
Rating            1
Reviews           0
Size              0
Installs          0
Type              0
Price             0
Content Rating    0
Genres            0
Last Updated      0
Current Ver       0
Android Ver       0
dtype: int64

Excel Missing Values:
Product     0
Price       0
Quantity    1
dtype: int64

Web Missing Values:
Country    0
Region     0
dtype: int64
```

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Code JupyterLab Python 3 (ipykernel)

```
2    87510  8.7M   5,000,000+  Free      0       Everyone
3   215644   25M  50,000,000+  Free      0           Teen
4      967  2.8M     100,000+  Free      0       Everyone

                    Genres      Last Updated         Current Ver  \
0             Art & Design   January 7, 2018               1.0.0
1  Art & Design;Pretend Play  January 15, 2018              2.0.0
2             Art & Design    August 1, 2018               1.2.4
3             Art & Design      June 8, 2018  Varies with device
4    Art & Design;Creativity     June 26, 2018              1.0.0

   Android Ver
0  4.0.3 and up
1  4.0.3 and up
2  4.0.3 and up
3    4.2 and up
4    4.4 and up

Processed Excel Data:
      Product  Price  Quantity
0      Laptop   1000       5.0
1  Smartphone    800       8.0
2      Tablet    500      15.0
3  Headphones    100      15.0
4    Keyboard     50      20.0

Processed Web Data:
    Country  Region
0   Algeria  AFRICA
1    Angola  AFRICA
2     Benin  AFRICA
3  Botswana  AFRICA
4   Burkina  AFRICA

=================================================================
8. SAVING PROCESSED CSV
=================================================================
Processed CSV saved successfully:
processed_text.csv

=================================================================
9. SAVING PROCESSED EXCEL
=================================================================
Processed Excel saved successfully:
processed_excel.xlsx

=================================================================
10. OUTPUT FILE VERIFICATION
=================================================================
✓ processed_text.csv created successfully.
✓ processed_excel.xlsx created successfully.

=================================================================
EXPERIMENT 2(C) COMPLETED SUCCESSFULLY
=================================================================
```